Matemàtiques
3r ESO

SOLUCIONS DE LA FITXA · UNITAT 5

Equacions

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Si no t'ix el mateix resultat, repassa els passos o consulta els apunts interactius de la unitat: hi trobaràs calculadores que resolen pas a pas qualsevol equació.

1 Equacions i identitats

1. ★☆☆

a) Identitat

b) Equació ($x = 3$)

c) Identitat

d) Equació ($x = \pm 1$)

e) Identitat

f) Equació ($x = 1$)

g) Identitat

h) Equació sense solució ($-2 \neq -1$)

2. ★☆☆

a) $4 = 4$: sí

b) $5 = 5$ i $5 = 5$: totes dues

c) $8 \neq 9$: no

d) $8 - 4 = 4$: sí

3. ★☆☆ 1r membre: $5x^2 - 3x + 1$; 2n membre: $2x - 7$; termes: $5x^2,\ -3x,\ 1,\ 2x,\ -7$; incògnita: x; grau 2.

4. ★★☆

a) $4x = -8$: grau 1

b) $-4x = 0$: grau 1

c) $-x = 2$: grau 1

d) $x^2 - 3x - 1 = 0$: grau 2

5. ★★☆

a) Les dues tenen la solució $x = 5$: sí.

b) La primera també té la solució $x = -5$: no.

6. ★★☆ Per exemple, $x + 2 = 0$ (o $3x + 1 = -5\ldots$) i $x^2 - 9 = 0$.

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7. ★★★

a) $18 - 3k + 3 = 0 \Rightarrow \boxed{k = 7}$

b) $-k - 2 = k - 6 \Rightarrow -2k = -4 \Rightarrow \boxed{k = 2}$

2 Equacions de primer grau

8. ★☆☆

a) $\boxed{x = 5}$

b) $\boxed{x = -7}$

c) $5x = 15 \Rightarrow \boxed{x = 3}$

d) $-2x = -8 \Rightarrow \boxed{x = 4}$

e) $2x = 8 \Rightarrow \boxed{x = 4}$

f) $-4x = -20 \Rightarrow \boxed{x = 5}$

g) $-3x = 12 \Rightarrow \boxed{x = -4}$

h) $-3x = 0 \Rightarrow \boxed{x = 0}$

9. ★☆☆

a) $3x - 6 = 12 \Rightarrow 3x = 18 \Rightarrow \boxed{x = 6}$

b) $5x + 5 = 2x - 7 \Rightarrow 3x = -12 \Rightarrow \boxed{x = -4}$

c) $-x + 1 = 2x - 5 \Rightarrow -3x = -6 \Rightarrow \boxed{x = 2}$

d) $6x - 2 = 4x + 8 \Rightarrow 2x = 10 \Rightarrow \boxed{x = 5}$

10. ★★☆

a) $4x - 11 = 5x - 13 \Rightarrow -x = -2 \Rightarrow \boxed{x = 2}$

b) $2x - 1 = 4x - 7 \Rightarrow -2x = -6 \Rightarrow \boxed{x = 3}$

c) $-2x + 3 = 7 \Rightarrow -2x = 4 \Rightarrow \boxed{x = -2}$

d) $3x - 7 = 2x + 2 \Rightarrow \boxed{x = 9}$

11. ★★☆

a) $2x + 6 = 2x + 6 \Rightarrow 0x = 0 \Rightarrow$ $\boxed{\text{infinites solucions}}$

b) $3x + 2 = 3x + 4 \Rightarrow 0x = 2 \Rightarrow$ $\boxed{\text{sense solució}}$

c) $-2 = -2 \Rightarrow 0x = 0 \Rightarrow$ $\boxed{\text{infinites solucions}}$

d) $x + 3 = 2x + 1 \Rightarrow -x = -2 \Rightarrow \boxed{x = 2}$

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12. ★★☆

a) $4x = 12 \Rightarrow \boxed{x = 3}$. Comprovació: $16 - 4 = 12$

b) $3x = 2x + 5 \Rightarrow \boxed{x = 5}$. Comprovació: $40 - 25 = 15 = 2 \cdot 5 + 5$

13. ★★★

a) $-4x + 2 = -7x + 8 \Rightarrow 3x = 6 \Rightarrow \boxed{x = 2}$

b) $2x - 10 = 2x + 2 \Rightarrow 0x = 12 \Rightarrow$ $\boxed{\text{sense solució}}$

14. ★★★ $3(5 - a) = 11 \Rightarrow 15 - 3a = 11 \Rightarrow \boxed{a = \frac{4}{3}}$

3 Equacions de primer grau amb denominadors

15. ★☆☆

a) m.c.m. $= 3$: $\boxed{x = 12}$

b) m.c.m. $= 4$: $2x + x = 24 \Rightarrow 3x = 24 \Rightarrow \boxed{x = 8}$

c) m.c.m. $= 3$: $3x - x = 30 \Rightarrow 2x = 30 \Rightarrow \boxed{x = 15}$

d) m.c.m. $= 5$: $2x - 5 = 15 \Rightarrow 2x = 20 \Rightarrow \boxed{x = 10}$

16. ★☆☆

a) m.c.m. $= 3$: $(x + 2) = 12 \Rightarrow x + 2 = 12 \Rightarrow \boxed{x = 10}$

b) m.c.m. $= 4$: $2(x - 1) = (x + 3) \Rightarrow 2x - 2 = x + 3 \Rightarrow \boxed{x = 5}$

c) m.c.m. $= 4$: $(3x - 1) = 8 \Rightarrow 3x - 1 = 8 \Rightarrow 3x = 9 \Rightarrow \boxed{x = 3}$

d) m.c.m. $= 12$: $2x + 12 = 3x \Rightarrow -x = -12 \Rightarrow \boxed{x = 12}$

17. ★★☆

a) m.c.m. $= 6$: $3(x + 1) - 2(x - 1) = 12 \Rightarrow x + 5 = 12 \Rightarrow \boxed{x = 7}$

b) m.c.m. $= 12$: $3(x - 2) - 2(x - 3) = 6 \Rightarrow \boxed{x = 6}$

c) m.c.m. $= 6$: $2(2x + 1) - 3(x - 2) = 18 \Rightarrow x + 8 = 18 \Rightarrow \boxed{x = 10}$

d) m.c.m. $= 10$: $10x - 2(x + 2) = 5(x + 6) \Rightarrow 8x - 4 = 5x + 30 \Rightarrow 3x = 34 \Rightarrow \boxed{x = \frac{34}{3}}$

18. ★★☆

a) m.c.m. $= 6$: $2(x - 1) + (x + 2) = 6x - 12 \Rightarrow 3x = 6x - 12 \Rightarrow -3x = -12 \Rightarrow \boxed{x = 4}$

b) m.c.m. $= 4$: $12 - (x - 1) = 2(x + 5) \Rightarrow -x + 13 = 2x + 10 \Rightarrow -3x = -3 \Rightarrow \boxed{x = 1}$

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19. ★★★

a) m.c.m. $= 12$: $9(x-2)-8(x+1)=2(x-10) \Rightarrow x-26=2x-20 \Rightarrow -x=6 \Rightarrow \boxed{x=-6}$

b) m.c.m. $= 12$: $6x-4(x-1)-3(x+1)=(1-x) \Rightarrow -x+1=-x+1 \Rightarrow 0x=0 \Rightarrow$ **infinites solucions**

c) m.c.m. $= 4$: $2(x+1)-(x+3)=x-4 \Rightarrow x-1=x-4 \Rightarrow 0x=-3 \Rightarrow$ **sense solució**

20. ★★★ El $-$ davant de la segona fracció afecta tot el numerador: $-3(x+1)=-3x-3$. $2x-4-3x-3=6 \Rightarrow -x=13 \Rightarrow \boxed{x=-13}$.

4 Equacions de segon grau incompletes

21. ★☆☆

a) $x^2-5x+6=0$: $\boxed{a=1,\ b=-5,\ c=6}$, completa

b) $3x^2-12=0$: $\boxed{a=3,\ b=0,\ c=-12}$, incompleta

c) $x^2-4x=0$: $\boxed{a=1,\ b=-4,\ c=0}$, incompleta

d) $x^2-4=0$: $\boxed{a=1,\ b=0,\ c=-4}$, incompleta

22. ★☆☆

a) $x^2=36 \Rightarrow \boxed{x=\pm 6}$

b) $x^2=25 \Rightarrow \boxed{x=\pm 5}$

c) $x^2=-9 \Rightarrow$ **sense solució**

d) $x^2=\frac{9}{4} \Rightarrow \boxed{x=\pm\frac{3}{2}}$

e) $x^2=9 \Rightarrow \boxed{x=\pm 3}$

f) $x^2=7 \Rightarrow \boxed{x=\pm\sqrt{7}}$

23. ★☆☆

a) $x\,(x-5)=0 \Rightarrow \boxed{x=0,\ x=5}$

b) $x^2+4x=0 \Rightarrow x\,(x+4)=0 \Rightarrow \boxed{x=-4,\ x=0}$

c) $2x^2-7x=0 \Rightarrow x\,(2x-7)=0 \Rightarrow \boxed{x=0,\ x=\frac{7}{2}}$

d) $x^2-6x=0 \Rightarrow x\,(x-6)=0 \Rightarrow \boxed{x=0,\ x=6}$

24. ★★☆

a) $x^2-9=7 \Rightarrow x^2=16 \Rightarrow \boxed{x=\pm 4}$

b) $x^2-2x=-2x+4 \Rightarrow x^2=4 \Rightarrow \boxed{x=\pm 2}$

c) $4x^2-4x+1=1 \Rightarrow x^2-x=0 \Rightarrow x\,(x-1)=0 \Rightarrow \boxed{x=0,\ x=1}$

d) $x^2+4x=3x \Rightarrow x^2+x=0 \Rightarrow x\,(x+1)=0 \Rightarrow \boxed{x=-1,\ x=0}$

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25. ★★☆

a) m.c.m. $= 3$: $x^2 = 36 \Rightarrow \boxed{x = \pm 6}$

b) m.c.m. $= 4$: $x^2 - 2x = 0 \Rightarrow x(x-2) = 0 \Rightarrow \boxed{x = 0,\ x = 2}$

c) m.c.m. $= 5$: $(x^2 + 1) = 10 \Rightarrow x^2 + 1 = 10 \Rightarrow x^2 = 9 \Rightarrow \boxed{x = \pm 3}$

d) m.c.m. $= 6$: $3(x^2 - 2) - 2(x^2 + 1) = 6 \Rightarrow x^2 - 8 = 6 \Rightarrow x^2 = 14 \Rightarrow \boxed{x = \pm\sqrt{14}}$

26. ★★★

a) $3x^2 - 2x - 10 = 2x^2 - 2x - 10 \Rightarrow x^2 = 0 \Rightarrow \boxed{x = 0 \text{ (doble)}}$

b) m.c.m. $= 6$: $3x(x-3) - 2x(x+1) = 6x \Rightarrow x^2 - 11x = 6x \Rightarrow x^2 - 17x = 0 \Rightarrow x(x-17) = 0 \Rightarrow \boxed{x = 0,\ x = 17}$

27. ★★★ En dividir entre x perd la solució $x = 0$. $5x^2 - 3x = 0 \Rightarrow x(5x-3) = 0 \Rightarrow \boxed{x = 0,\ x = \frac{3}{5}}$.

28. ★★★

a) P. ex. $\boxed{x^2 - 16 = 0}$

b) P. ex. $\boxed{x^2 - 3x = 0}$

5 Equacions de segon grau completes

29. ★☆☆

a) $\Delta = 4 \Rightarrow x = \frac{6 \pm 2}{2} \Rightarrow \boxed{x = 2,\ x = 4}$

b) $\Delta = 36 \Rightarrow x = \frac{-2 \pm 6}{2} \Rightarrow \boxed{x = -4,\ x = 2}$

c) $\Delta = 36 \Rightarrow x = \frac{4 \pm 6}{2} \Rightarrow \boxed{x = -1,\ x = 5}$

d) $\Delta = 1 \Rightarrow x = \frac{-7 \pm 1}{2} \Rightarrow \boxed{x = -4,\ x = -3}$

30. ★☆☆

a) $\Delta = 9 \Rightarrow x = \frac{5 \pm 3}{4} \Rightarrow \boxed{x = \frac{1}{2},\ x = 2}$

b) $\Delta = 25 \Rightarrow x = \frac{-1 \pm 5}{6} \Rightarrow \boxed{x = -1,\ x = \frac{2}{3}}$

c) $\Delta = 25 \Rightarrow x = \frac{1 \pm 5}{12} \Rightarrow \boxed{x = -\frac{1}{3},\ x = \frac{1}{2}}$

d) $\Delta = 64 \Rightarrow x = \frac{-4 \pm 8}{8} \Rightarrow \boxed{x = -\frac{3}{2},\ x = \frac{1}{2}}$

31. ★☆☆

a) $\Delta = 0$: $\boxed{\text{una solució}}$

b) $\Delta = -23$: $\boxed{\text{cap solució}}$

c) $\Delta = 5$: $\boxed{\text{dues solucions}}$

d) $\Delta = 16$: $\boxed{\text{dues solucions}}$

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32. ★★☆

a) $x^2 - x - 6 = 0 \Rightarrow \Delta = 25 \Rightarrow x = \frac{1 \pm 5}{2} \Rightarrow \boxed{x = -2,\ x = 3}$

b) $\Delta = 0 \Rightarrow x = \frac{6}{18} \Rightarrow \boxed{x = \frac{1}{3}\ \text{(doble)}}$

c) $\Delta = -11 \Rightarrow \boxed{\text{sense solució}}$

d) $\Delta = 12 \Rightarrow x = \frac{2 \pm \sqrt{12}}{2} \Rightarrow \boxed{x = 1 \pm \sqrt{3}}$

33. ★★☆

a) $x^2 + 3x - 4 = 6 \Rightarrow x^2 + 3x - 10 = 0 \Rightarrow \Delta = 49 \Rightarrow x = \frac{-3 \pm 7}{2} \Rightarrow \boxed{x = -5,\ x = 2}$

b) $x^2 + 4x + 4 = 3x + 10 \Rightarrow x^2 + x - 6 = 0 \Rightarrow \Delta = 25 \Rightarrow x = \frac{-1 \pm 5}{2} \Rightarrow \boxed{x = -3,\ x = 2}$

c) $2x^2 - x = x^2 + 2x - 3 \Rightarrow x^2 - 3x + 3 = 0 \Rightarrow \Delta = -3 \Rightarrow \boxed{\text{sense solució}}$

d) $2x^2 - 6x + 1 = 9 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow \Delta = 25 \Rightarrow x = \frac{3 \pm 5}{2} \Rightarrow \boxed{x = -1,\ x = 4}$

34. ★★☆

a) m.c.m. $= 6$: $3x^2 - 2x = 8 \Rightarrow 3x^2 - 2x - 8 = 0 \Rightarrow \Delta = 100 \Rightarrow x = \frac{2 \pm 10}{6} \Rightarrow \boxed{x = -\frac{4}{3},\ x = 2}$

b) m.c.m. $= 6$: $2(x^2 - 1) + 3x = 12 \Rightarrow 2x^2 + 3x - 2 = 12 \Rightarrow 2x^2 + 3x - 14 = 0 \Rightarrow \Delta = 121 \Rightarrow x = \frac{-3 \pm 11}{4} \Rightarrow \boxed{x = -\frac{7}{2},\ x = 2}$

35. ★★★

a) m.c.m. $= 12$: $3(x-1)^2 - 2(x+2) = 4(x-3) \Rightarrow 3x^2 - 8x - 1 = 4x - 12 \Rightarrow 3x^2 - 12x + 11 = 0 \Rightarrow \Delta = 12 \Rightarrow x = \frac{12 \pm \sqrt{12}}{6} \Rightarrow \boxed{x = \frac{6 \pm \sqrt{3}}{3}}$

b) m.c.m. $= 6$: $3x(x+1) - 2(x-1)(x+2) = 6 \Rightarrow x^2 + x + 4 = 6 \Rightarrow x^2 + x - 2 = 0 \Rightarrow \Delta = 9 \Rightarrow x = \frac{-1 \pm 3}{2} \Rightarrow \boxed{x = -2,\ x = 1}$

36. ★★★

a) $\Delta = 16 - 4k = 0 \Rightarrow \boxed{k = 4}$; $x^2 - 4x + 4 = 0 \Rightarrow \boxed{x = 2}$ (doble)

b) $9 + 3k - 6 = 0 \Rightarrow \boxed{k = -1}$; $x^2 - x - 6 = 0 \Rightarrow x = 3, \boxed{x = -2}$

37. ★★★

a) $(x-2)(x+5) = 0 \Rightarrow \boxed{x^2 + 3x - 10 = 0}$

b) $(2x-1)(x-3) = 0 \Rightarrow \boxed{2x^2 - 7x + 3 = 0}$

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6 Equacions de tercer grau (Ruffini)

38. ★☆☆

a) $\boxed{x=2,\ x=-3,\ x=5}$

b) $\boxed{x=0,\ x=-1,\ x=\frac{3}{2}}$

c) $\boxed{x=4,\ x=3,\ x=-3}$

d) $x^2+4=0$ no té solució: $\boxed{x=-1}$

39. ★☆☆

a) $x\left(x^2-16\right)=0 \Rightarrow x^2=16 \Rightarrow \boxed{x=\pm 4,\ x=0}$

b) $x\left(x^2+3x-10\right)=0 \Rightarrow \Delta=49 \Rightarrow x=\frac{-3\pm 7}{2} \Rightarrow \boxed{x=-5,\ x=0,\ x=2}$

c) $x^3-4x^2=0 \Rightarrow x^2(x-4)=0 \Rightarrow \boxed{x=0\ (\text{doble}),\ x=4}$

d) $x\left(x^2+x+1\right)=0 \Rightarrow \Delta=-3 \Rightarrow \boxed{x=0}$

40. ★★☆

a)
$$\begin{array}{c|ccc|c} & 1 & 0 & -7 & 6 \\ 1 & & 1 & 1 & -6 \\ \hline & 1 & 1 & -6 & 0 \end{array} \Rightarrow (x-1)(x^2+x-6)=0 \Rightarrow \Delta=25 \Rightarrow x=\frac{-1\pm 5}{2} \Rightarrow \boxed{x=-3,\ x=1,\ x=2}$$

b)
$$\begin{array}{c|ccc|c} & 1 & -4 & 1 & 6 \\ -1 & & -1 & 5 & -6 \\ \hline & 1 & -5 & 6 & 0 \end{array} \Rightarrow (x+1)(x^2-5x+6)=0 \Rightarrow \Delta=1 \Rightarrow x=\frac{5\pm 1}{2} \Rightarrow \boxed{x=-1,\ x=2,\ x=3}$$

c)
$$\begin{array}{c|ccc|c} & 1 & 2 & -5 & -6 \\ -1 & & -1 & -1 & 6 \\ \hline & 1 & 1 & -6 & 0 \end{array} \Rightarrow (x+1)(x^2+x-6)=0 \Rightarrow \Delta=25 \Rightarrow x=\frac{-1\pm 5}{2} \Rightarrow \boxed{x=-3,\ x=-1,\ x=2}$$

d)
$$\begin{array}{c|ccc|c} & 1 & -3 & -4 & 12 \\ 2 & & 2 & -2 & -12 \\ \hline & 1 & -1 & -6 & 0 \end{array} \Rightarrow (x-2)(x^2-x-6)=0 \Rightarrow \Delta=25 \Rightarrow x=\frac{1\pm 5}{2} \Rightarrow \boxed{x=\pm 2,\ x=3}$$

41. ★★☆

a)
$$\begin{array}{c|ccc|c} & 1 & -1 & -8 & 12 \\ 2 & & 2 & 2 & -12 \\ \hline & 1 & 1 & -6 & 0 \end{array} \Rightarrow (x-2)(x^2+x-6)=0 \Rightarrow \Delta=25 \Rightarrow x=\frac{-1\pm 5}{2} \Rightarrow \boxed{x=-3,\ x=2\ (\text{doble})}$$

b)
$$\begin{array}{c|ccc|c} & 1 & 3 & 3 & 1 \\ -1 & & -1 & -2 & -1 \\ \hline & 1 & 2 & 1 & 0 \end{array} \Rightarrow (x+1)(x^2+2x+1)=0 \Rightarrow \Delta=0 \Rightarrow x=\frac{-2}{2} \Rightarrow \boxed{x=-1\ (\text{triple})}$$

c)
$$\begin{array}{c|ccc|c} & 1 & -2 & 1 & -2 \\ 2 & & 2 & 0 & 2 \\ \hline & 1 & 0 & 1 & 0 \end{array} \Rightarrow (x-2)(x^2+1)=0 \Rightarrow x^2=-1 \Rightarrow \boxed{x=2}$$

d)
$$\begin{array}{c|ccc|c} & 1 & 1 & 0 & -2 \\ 1 & & 1 & 2 & 2 \\ \hline & 1 & 2 & 2 & 0 \end{array} \Rightarrow (x-1)(x^2+2x+2)=0 \Rightarrow \Delta=-4 \Rightarrow \boxed{x=1}$$

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42. ★★☆

a) $x^3-4x^2-3x+12=0 \Rightarrow$

	1	−4	−3	12
4		4	0	−12
	1	0	−3	0

$\Rightarrow (x-4)(x^2-3)=0 \Rightarrow x^2=3 \Rightarrow$ $\boxed{x=4,\ x=\pm\sqrt{3}}$

b) $x^3-x=24 \Rightarrow x^3-x-24=0 \Rightarrow$

	1	0	−1	−24
3		3	9	24
	1	3	8	0

$\Rightarrow (x-3)(x^2+3x+8)=0 \Rightarrow \Delta=-23 \Rightarrow$ $\boxed{x=3}$

43. ★★★

a)

	2	−5	−4	3
−1		−2	7	−3
	2	−7	3	0

$\Rightarrow (x+1)(2x^2-7x+3)=0 \Rightarrow \Delta=25 \Rightarrow x=\frac{7\pm5}{4} \Rightarrow$ $\boxed{x=-1,\ x=\frac{1}{2},\ x=3}$

b)

	3	4	−5	−2
1		3	7	2
	3	7	2	0

$\Rightarrow (x-1)(3x^2+7x+2)=0 \Rightarrow \Delta=25 \Rightarrow x=\frac{-7\pm5}{6} \Rightarrow$ $\boxed{x=-2,\ x=-\frac{1}{3},\ x=1}$

44. ★★★

a)

	1	0	−10	0	9
1		1	1	−9	−9
	1	1	−9	−9	0

$\Rightarrow (x-1)(x^3+x^2-9x-9)=0 \Rightarrow$

	1	1	−9	−9
−1		−1	0	9
	1	0	−9	0

$\Rightarrow (x+1)(x^2-9)=0 \Rightarrow x^2=9 \Rightarrow$ $\boxed{x=\pm3,\ x=\pm1}$

b)

	1	−1	−7	1	6
1		1	0	−7	−6
	1	0	−7	−6	0

$\Rightarrow (x-1)(x^3-7x-6)=0 \Rightarrow$

	1	0	−7	−6
−1		−1	1	6
	1	−1	−6	0

$\Rightarrow (x+1)(x^2-x-6)=0 \Rightarrow \Delta=25 \Rightarrow x=\frac{1\pm5}{2} \Rightarrow$ $\boxed{x=-2,\ x=\pm1,\ x=3}$

45. ★★★ $8-4k+4+4=0 \Rightarrow \boxed{k=4}$.

	1	−4	2	4
2		2	−4	−4
	1	−2	−2	0

$x^2-2x-2=0 \Rightarrow \boxed{x=1\pm\sqrt{3}}$

7 Problemes amb equacions

46. ★☆☆ $x+(x+1)=75 \Rightarrow 2x=74 \Rightarrow x=37$: 37 i 38

47. ★☆☆ $x+3x=56 \Rightarrow 4x=56 \Rightarrow \boxed{x=14}$

48. ★☆☆ $x-\frac{x}{3}-\frac{x}{4}=25 \xrightarrow{\cdot 12} 12x-4x-3x=300 \Rightarrow 5x=300 \Rightarrow$ 60 €

49. ★☆☆ $40+x=2(12+x) \Rightarrow 40+x=24+2x \Rightarrow \boxed{x=16}$ anys

50. ★★☆ $2x+2(2x+3)=54 \Rightarrow 6x+6=54 \Rightarrow x=8$: 8 m i 19 m

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51. ★★☆ Gallines x: $2x + 4(40 - x) = 110 \Rightarrow -2x = -50 \Rightarrow x = 25$: **25 gallines i 15 conills**

52. ★★☆ Monedes de 2 €: x. $2x + (30 - x) = 46 \Rightarrow \boxed{x = 16}$: **16 de 2 € i 14 d'1 €**

53. ★★☆ $3x + 12 = 2(x + 12) \Rightarrow \boxed{x = 12}$: fill **12 anys**, mare **36 anys**

54. ★★☆ $x(x + 1) = 156 \Rightarrow x^2 + x - 156 = 0 \Rightarrow x = \frac{-1 \pm 25}{2}$: $x = 12$ (-13 no és natural): **12 i 13**

55. ★★☆ $x^2 - 3x - 40 = 0 \Rightarrow x = \frac{3 \pm 13}{2}$: $\boxed{x = 8 \text{ o } x = -5}$

56. ★★☆ $(x + 3)^2 - x^2 = 51 \Rightarrow 6x + 9 = 51 \Rightarrow \boxed{x = 7 \text{ cm}}$

57. ★★★ $x^2 + (x + 2)^2 = 100 \Rightarrow x^2 + 2x - 48 = 0 \Rightarrow x = \frac{-2 \pm 14}{2}$: $x = 6$ (-8 no val): **6 cm i 8 cm**

58. ★★★ Si un costat fa x, l'altre fa $17 - x$: $x(17 - x) = 60 \Rightarrow x^2 - 17x + 60 = 0 \Rightarrow x = \frac{17 \pm 7}{2}$: **12 m i 5 m**

59. ★★★ $12x + 8(20 - x) = 9{,}5 \cdot 20 \Rightarrow 4x + 160 = 190 \Rightarrow x = 7{,}5$: **7,5 kg de 12 € i 12,5 kg de 8 €**

60. ★★★ $20t + 25t = 90 \Rightarrow \boxed{t = 2 \text{ h}}$; a $20 \cdot 2 = \boxed{40 \text{ km}}$

61. ★★★ $x^2(x + 3) = 112 \Rightarrow x^3 + 3x^2 - 112 = 0$.

	1	3	0	−112
4		4	28	112
	1	7	28	0

($x^2 + 7x + 28 = 0$ no té solució): **base de 4 cm × 4 cm i altura 7 cm**

62. ★★★ $x(x + 1)(x + 2) = 8(3x + 3) \Rightarrow x^3 + 3x^2 - 22x - 24 = 0 \Rightarrow (x + 1)(x^2 + 2x - 24) = 0$: $x = -1, 4, -6$; només $x = 4$ és natural: **4, 5 i 6**

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